

Online Complaint Registration and Management System

Phase 6: Project Documentation

SmartBridge Internship Project Submission

Team: Online Complaint Registration and Management System

6.1 Introduction

The Online Complaint Registration and Management System is a full-stack MERN web application that allows citizens/customers to register complaints, track their resolution status in real time, and communicate implicitly through status and agent updates, while agents and admins manage the resolution workflow end-to-end. This document consolidates the outcomes of all project phases: Ideation, Requirement Analysis, Design, Planning, Development, and Testing.

6.2 Project Overview

Attribute	Details
Project Title	Online Complaint Registration and Management System
Domain	Web Application — Governance / Customer Service
Tech Stack	React.js, Node.js, Express.js, MongoDB (MERN)
User Roles	User, Agent, Admin
Hosting	Render (Static Site + Web Service)
Database	MongoDB Atlas

6.3 Feature Summary

- Secure registration and login (JWT + bcrypt password hashing)
- Role-based dashboards for Users, Agents, and Admins
- Complaint registration with category, location, and priority
- Real-time status tracking (Pending → Assigned → In Progress → Resolved → Closed)
- Agent assignment by Admin
- Automatic notifications on complaint events
- Search and filter complaints by status, category, or keyword
- Admin analytics dashboard (totals, status/category breakdown)
- User management (role changes, account deletion) by Admin

6.4 Performance Testing

The following test cases were executed to validate the core functionality before final submission. (Fill in Actual Result / Status after running each test on the deployed or local build.)

Test Case	Steps	Expected Result	Actual Result / Status
User Registration	Register with valid name, email, password	Account created; JWT returned; redirected to dashboard	
Duplicate Registration	Register with an already-used email	Error: "User already exists with this email"	
Login — valid credentials	Log in with correct email/password	Logged in; redirected based on role	
Login — invalid credentials	Log in with wrong password	Error: "Invalid email or password"	

Test Case	Steps	Expected Result	Actual Result / Status
Complaint Submission	Submit complaint with all required fields	Complaint created with status "Pending"; notification generated	
Complaint Tracking	View Dashboard as the submitting user	Only own complaints are listed	
Agent Assignment	Admin assigns an agent to a complaint	Status auto-updates to "Assigned"; both user and agent notified	
Status Update	Agent updates status to "In Progress"	Status updates; user notified of the change	
Unauthorized Access	Non-admin attempts to access /api/complaints/stats	Request rejected with 403 Forbidden	
Role Management	Admin changes a user's role to "agent"	Role updated; user gains agent-level access on next login	

6.5 Advantages

- Centralized, transparent complaint tracking replacing manual/phone-based processes
- Role-based access ensures accountability at every stage
- Real-time notifications keep users informed without manual follow-up
- Analytics dashboard gives admins visibility into complaint trends and team performance

6.6 Limitations

- No real-time push notifications (currently notification records are fetched on demand, not pushed live via WebSockets)
- No file/image attachment support for complaints in the current version
- No SMS/email delivery of notifications — in-app only

6.7 Future Scope

- Integrate WebSockets (Socket.IO) for live notification delivery
- Add file/photo upload support for complaint evidence
- Add email/SMS notification channels
- Add a public complaint-heatmap view for transparency
- Add SLA-based auto-escalation for overdue complaints

6.8 Conclusion

The Online Complaint Registration and Management System successfully addresses the core problem identified during the Ideation phase — the lack of a centralized, transparent complaint-handling process — by delivering a secure, role-based, full-stack web application with real-time tracking and administrative oversight. The system was designed, developed, tested, and deployed following an iterative sprint-based approach, and is ready for demonstration and evaluation.